

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
MERN Stack

M2:Practical – VII

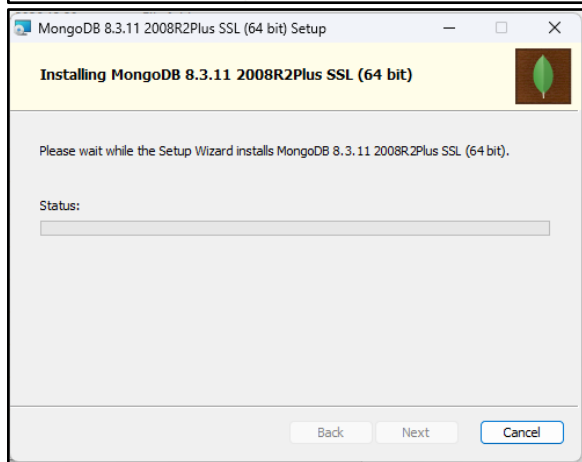
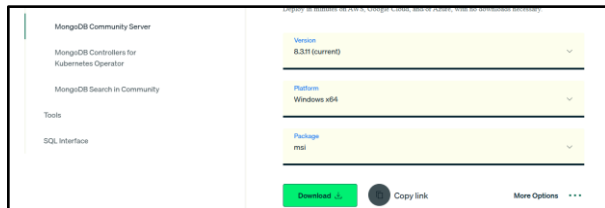
Roll No.: T002	Name: Asmitha Mohan Raj
Class: TYIT	Batch: 1
Date of Assignment: 19/09/2026	Date/Time of Submission: 19/09/2026

Aim: MongoDB Basics.

A. Install and start MongoDB.

Step:

- Go to the official MongoDB website and download MongoDB Community Server for Windows:
- [MongoDB Community Server Download](#)
- Choose:
 - Version: Current/available version
 - Platform: Windows
 - Package: MSI
- Download the .msi installer.

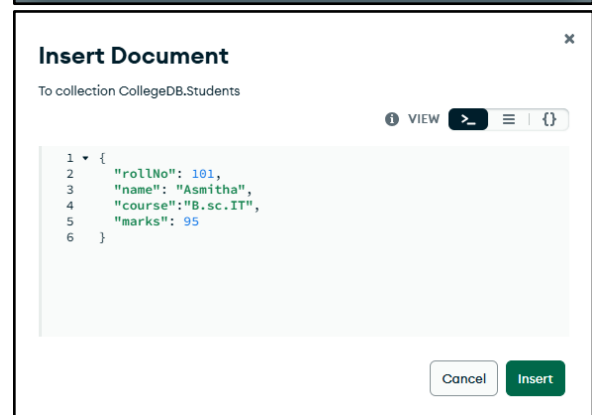
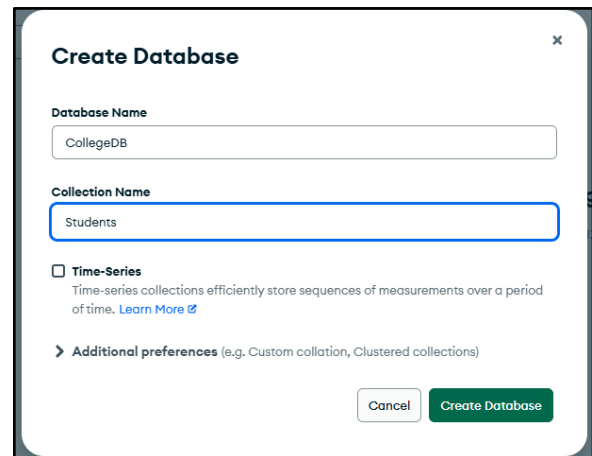


B. Create a database and collection.

Step:

- Click + symbol beside localhost, it is create database

- Enter Database Name - CollegeDB , collection name - Students
- After create database
- Now click ADD DATA -> Insert document



C. Insert records in MongoDB.

Step:

- “Open MongoDB Shell” button at the top-right, click: Open MongoDB Shell
- Command to choose database, CollegeDB prompt will start.

```
> _MONGOSH
> use("CollegeDB")
< switched to db CollegeDB
> db["Students"].find()
< {
  _id: ObjectId('6aae2a63f34e548a6db6da01'),
  rollNo: 101,
  name: 'Asmitha',
  course: 'B.sc.IT',
  marks: 95
}
```

3. insertOne() → inserts one document
4. After enter it will acknowledged

```
> db.students.insertOne({})
< {
  acknowledged: true,
  insertedId: ObjectId('6aae2b2b2664e0e144f14712')
}
```

5. insertMany() → inserts multiple documents
6. After enter

```
> db.students.insertMany([{rollNo: 102, name: "Anu", course: "IT", marks: 100}, {rollNo: 103}])
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6aae2ce82664e0e144f14714'),
    '1': ObjectId('6aae2ce82664e0e144f14715')
  }
}
```

D. View records from the collection.

Step:

1. db.students.find() - View all records. This displays all student documents.
2. db.students.find().pretty() - View records in a readable format
3. In mongosh window both outputs will be same.
- 4.

```
db.students.find()
{
  _id: ObjectId('6aae2b2b2664e0e144f14712')
}
{
  _id: ObjectId('6aae2c0a2664e0e144f14713'),
  rollNo: 102
}
```

```
{
  _id: ObjectId('6aae2ce82664e0e144f14714'),
  rollNo: 102,
  name: 'Anu',
  course: 'IT',
  marks: 100
}
{
  _id: ObjectId('6aae2ce82664e0e144f14715'),
  rollNo: 103,
  name: 'Ammu',
  course: 'IT',
  marks: 85
}
```

5. Find a particular student - find student with roll number 101:

```
> db.students.find({ rollNo: 102})
< {
  _id: ObjectId('6aae2c0a2664e0e144f14713'),
  rollNo: 102
}
{
  _id: ObjectId('6aae2ce82664e0e144f14714'),
  rollNo: 102,
  name: 'Anu',
  course: 'IT',
  marks: 100
}
```

```
> db.students.find({ marks: 85})
< {
  _id: ObjectId('6aae2ce82664e0e144f14715'),
  rollNo: 103,
  name: 'Ammu',
  course: 'IT',
  marks: 85
}
```

6. You can find anything from the table.

E. Update and delete records.

Step:

1. Delete - Delete the student with roll number 104.
2. deleteOne - deletes only 1 document
3. deleteMany - deletes all matching documents

```
> db.students.deleteOne({rollNo: 103})
< {
  acknowledged: true,
  deletedCount: 1
}
```

4. Update - course need to be changed from "IT" to "CS".
5. updateOne - updates only 1 document
6. updateMany - updates all matching documents

```
> db.students.updateOne({rollNo: 102},{set: {marks: 95}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

Output:

```
> db.students.find()
< {
  _id: ObjectId('6aae2b2b2664e0e144f14712')
}
{
  _id: ObjectId('6aae2c0a2664e0e144f14713'),
  rollNo: 102,
  marks: 95
}
{
  _id: ObjectId('6aae2ce82664e0e144f14714'),
  rollNo: 102,
  name: 'Anu',
  course: 'IT',
  marks: 100
}
```

F. Connect MongoDB with Node.js.

Steps:

1. Open folder in vs code
2. Run npm init -y to create package.json file.
3. Add server.js file
4. Run npm install mongoose
5. Mongoose is a Node.js library that helps Node.js communicate with
6. MongoDB.
7. Add following code into server.js file

Code:

```
const mongoose = require("mongoose");

mongoose.connect("mongodb://127.0.0.1:27017/CollegeD
B")
.then(() => {
  console.log("MongoDB Connected Successfully");
})
.catch((error) => {
  console.log("MongoDB Connection Failed");
  console.log(error);
});
```

```
});
```

Output:

```
PS C:\Users\itlab\Desktop\Asmitha TYIT\MongoDB Prac-7> node server.js
MongoDB Connected Successfully
```

8. To view in terminal.

Code:

```
const mongoose = require("mongoose");
mongoose.connect("mongodb://127.0.0.1:27017/CollegeD
B")
.then(async () => {
  console.log("MongoDB Connected Successfully");

  const students = await mongoose.connection.db
    .collection("students")
    .find()
    .toArray();

  console.log(students);
})
.catch((error) => {
  console.log("MongoDB Connection Failed");
  console.log(error);
});
```

Output:

```
PS C:\Users\itlab\Desktop\Asmitha TYIT\MongoDB Prac-7> node server.js
MongoDB Connected Successfully
[
  { _id: new ObjectId('6aae2b2b2664e0e144f14712') },
  {
    _id: new ObjectId('6aae2c0a2664e0e144f14713'),
    rollNo: 102,
    marks: 95
  },
  {
    _id: new ObjectId('6aae2ce82664e0e144f14714'),
    rollNo: 102,
    name: 'Anu',
    course: 'IT',
    marks: 100
  }
]
```